Fist Order Metric Reconstruction in Schwarzschld spacetime

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Abstract

My notes!

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CONTENTS

I. GHP & HELD

Combining eq. 123, eq. 124 and eq. J.1a of [1] gives

$$\zeta = -\Psi_2^{-1/6}\bar{\Psi}_2^{-1/6}\rho^{-1/2}\bar{\rho}^{1/2} \tag{1}$$

$$\begin{aligned} B_a &= -\rho n_a + \tau \bar{m}_a, \\ B'_a &= -\rho' l_a + \tau' m_a. \end{aligned} \tag{2}$$

$$t_a = \zeta(B_a - B'_a) \tag{3}$$

$$Ł_t\eta = t^a\Theta_a\eta - \frac{p}{2}\zeta\Psi_2\eta - \frac{q}{2}\bar{\zeta}\bar{\Psi}_2\eta = \zeta\rho'Þ\eta - \zeta\rhoÞ'\eta - \zeta\tau'ð\eta + \zeta\tauð''\eta - \frac{p}{2}\zeta\Psi_2\eta - \frac{q}{2}\bar{\zeta}\bar{\Psi}_2\eta \tag{4}$$

$$\begin{aligned} ÞÞÞÞ\left(\zeta^4\psi_4\right) &= ð'ð'ð'ð'\left(\zeta^4\psi_0\right) - 3Ł_t\bar{\psi}_0, \\ Þ'Þ'Þ'Þ'\left(\zeta^4\psi_0\right) &= ðððð\left(\zeta^4\psi_4\right) + 3Ł_t\bar{\psi}_4. \end{aligned} \tag{5}$$

II. COMPONENTS OF HOMOGENOUS Φ

In this section, I express how one achieves a metric perturbation that falls fast enough at $\mathscr{I}^+$. To achieve this end I used the procedure demonstrated in details by Hollands et. al. in [1]. This method is based on the GHZ reconstruction [2] combined with the puncture scheme [3]. The puncture scheme implemented in [3] breaks the retarded solution $h_{ab}^{\rm ret}$ into two parts of punctured field $h_{ab}^{\rm P}$ which contains the singular field field and the residual part $h_{ab}^{\rm Res}$. The residual field satisfy a homogeneous linearized Einstein equation sourced by $\mathcal{E}h_{ab}^{\rm P}$ away from the singular source, i.e.,

$$\mathcal{E}h_{ab}^{\rm Res} = -\mathcal{E}h_{ab}^{\rm P}. \tag{6}$$

The puncture field is an estimate for the singular field in a neighborhood of the orbiting compact object. This estimate is given by the product of the singular field and a $W(r, \theta, \phi)$ with the following properties:

1. W is compactly supported (usually $W = 0$ for $r < r_{\rm min}$ and $r > r_{\rm max}$, the support of W is called puncture region;
2. $W = 1$ on the location of the compact object.

One obvious choice would be $W(r) = \Theta(r - r_{\rm min}) - \Theta(r_{\rm max} - r)$ but one might choose a smooth one too. Consequently, residual field $h_{ab}^{\rm Res}$ is only sourced in the puncture region.

Applying $\mathcal{S}$ on both sides of (6) and using the Wald identity [**?**],

$$\mathcal{O}\psi_0^{\rm Res} = -\mathcal{S}^{ab}T_{ab}^{\rm P} = -T_0^{\rm P}. \tag{7}$$

Using the Wald identity [**?**], we have

Combining eq. 123, eq. 124 and eq. J.1a of [1] gives

$$2\tilde{\text{Þ}}'\tilde{\text{Þ}}'\tilde{\text{Þ}}'\bar{\psi}_0^{5\circ} = \tilde{\eth}'\tilde{\eth}'\tilde{\eth}'\tilde{\eth}'\tilde{\eth}\tilde{\eth}\tilde{\eth}\tilde{\eth}\bar{f}^\circ - 9\Psi^\circ\bar{\Psi}^\circ\tilde{\text{Þ}}'\tilde{\text{Þ}}'\bar{f}^\circ. \tag{8}$$

By expanding $\bar{f}^\circ$ and $\bar{\psi}_0^{5\circ}$ in spheroidal (spin-weighted harmonics in Schwarzschild) harmonics, one can reverse this equation and obtain $\bar{f}^\circ$. Next, we obtain foloowing Held-like scalars $\Phi^{i\circ}$ from $\bar{f}^\circ$ (and potentially ODEs in terms of $\frac{\partial}{\partial u}$).

$$\tilde{\text{Þ}}'\tilde{\text{Þ}}'\Phi^{0\circ} = -\frac{1}{3}\tilde{\eth}'\tilde{\eth}'\tilde{\eth}'\tilde{\eth}\tilde{\eth}\tilde{\eth}\bar{f}^\circ - \left[2\bar{\tau}^\circ\tilde{\eth}'\tilde{\eth}\tilde{\eth} + \Psi^\circ\tilde{\eth}'\tilde{\eth} + 2\Psi^\circ\rho'^\circ\right]\tilde{\text{Þ}}'\bar{f}^\circ \tag{9}$$

$$\tilde{\text{Þ}}'\Phi^{1\circ} = -\tilde{\eth}'\tilde{\eth}'\tilde{\eth}\tilde{\eth}\bar{f}^\circ - \left[4\bar{\tau}^\circ\tilde{\eth} + 3\Psi^\circ\right]\tilde{\text{Þ}}'\bar{f}^\circ \tag{10}$$

$$\Phi^{2\circ} = -2\tilde{\eth}'\tilde{\eth}\bar{f}^\circ, \tag{11}$$

$$\Phi^{3\circ} = -2\tilde{\text{Þ}}'\bar{f}^\circ, \tag{12}$$

$$\bar{\zeta}_m^\circ = \tilde{\eth}\bar{f}^\circ, \tag{13}$$

$$\tilde{\text{Þ}}'\zeta_l^\circ = -\operatorname{Re}\left\{\tilde{\eth}\tilde{\eth}\bar{f}^\circ\right\}, \tag{14}$$

$$\tilde{\text{Þ}}'\zeta_n^\circ = -\operatorname{Re}\left\{\tilde{\eth}'\tilde{\eth}\tilde{\eth}\tilde{\eth}\bar{f}^\circ - \rho'^\circ\tilde{\eth}\tilde{\eth}\bar{f}^\circ - 2\tau^\circ\tilde{\eth}\tilde{\text{Þ}}'\bar{f}^\circ - \frac{1}{2}\Omega^\circ\tilde{\eth}\tilde{\eth}\tilde{\text{Þ}}'\bar{f}^\circ\right\}, \tag{15}$$

kslfhadlf

$$\begin{aligned} h_{ab} &\to h_{ab} - \mathcal{L}_\zeta g_{ab} \\ \Phi &\to \Phi + \Phi^{0\circ} + \Phi^{1\circ}\frac{1}{\rho} + \Phi^{2\circ}\frac{1}{\rho^2} + \Phi^{3\circ}\frac{1}{\rho^3} \end{aligned} \tag{16}$$

$$\begin{aligned} \mathcal{L}_\zeta g_{ab} l^a &= 0 \\ \mathcal{L}_\zeta g_{ab} m^a \bar{m}^b &= 0 \end{aligned} \tag{17}$$

$$
\mathcal{L}_\zeta g_{ab} n^a n^b = \mathrm{Re}\Bigg[4\rho\bar\rho\bar\tau^\circ \tilde{\eth}'\tilde{\eth}\tilde{\eth}\xi_l^\circ - 4\rho^2\bar\rho\bar\tau^{\circ 2}\tilde{\eth}\tilde{\eth}\xi_l^\circ + 2\rho^2\Psi^\circ\tilde{\eth}'\tilde{\eth}\xi_l^\circ + 4\rho^2\bar\rho\Psi^\circ\bar\tau^\circ\tilde{\eth}\xi_l^\circ
$$

$$
-2\tilde{\eth}'\tilde{\eth}'\tilde{\eth}'\tilde{\eth} f^\circ - 2\left(\frac{1}{\rho} + 4\Omega^\circ\right)\tilde{\eth}'\tilde{\eth}'\tilde{\text{Þ}}' f^\circ - 4\rho\bar\rho\Omega^\circ\bar\tau^\circ\tilde{\eth}'\tilde{\eth}'\tilde{\eth} f^\circ + 4\left(4 + \bar\rho\Omega^\circ - \rho\bar\rho\Omega^{\circ 2}\right)\bar\tau^\circ\tilde{\eth}'\tilde{\text{Þ}}' f^\circ
$$

$$
-\left(8\rho'^\circ + 3\rho\Psi^\circ + 3\bar\rho\bar\Psi^\circ + 2\rho^2\Omega^{\circ 2}\Psi^\circ + 8\rho\bar\rho\tau^\circ\bar\tau^\circ\right)\tilde{\eth}'\tilde{\eth}' f^\circ + 4\rho^2\bar\tau^{\circ 2}\tilde{\eth}'\tilde{\eth} f^\circ
$$

$$
+4\rho\left(\rho + 2\bar\rho\right)\Omega^\circ\bar\tau^{\circ 2}\tilde{\text{Þ}}' f^\circ + 8\rho^2\bar\rho\tau^\circ\bar\tau^{\circ 2}\tilde{\eth}' f^\circ + 4\rho\bar\rho\bar\Psi^\circ\bar\tau^\circ\tilde{\eth}' f^\circ - 24\rho\bar\rho\Omega^\circ\rho'^\circ\bar\tau^\circ\tilde{\eth}' f^\circ + 16\rho^2\rho'^\circ\bar\tau^{\circ 2} f^\circ\Bigg] \tag{18}
$$

$$
\mathcal{L}_\zeta g_{ab} n^a m^b = \bar\rho\tilde{\eth}'\tilde{\eth}\tilde{\eth}\xi_l^\circ - 2\rho\bar\rho\bar\tau\tilde{\eth}\tilde{\eth}\xi_l^\circ + \rho(\rho+\bar\rho)\Psi^\circ\tilde{\eth}\xi_l^\circ - \bar\rho\Omega^\circ\tilde{\eth}'\tilde{\eth}'\tilde{\eth} f^\circ + 2\rho\bar\tau^\circ\tilde{\eth}'\tilde{\eth} f^\circ
+ \left(\frac{1}{\rho} + \frac{\bar\rho}{\rho}\Omega^\circ - \bar\rho\Omega^{\circ 2}\right)\tilde{\eth}'\tilde{\text{Þ}}' f^\circ + 2(\rho+\bar\rho)\Omega^\circ\bar\tau^\circ\tilde{\text{Þ}}' f^\circ - 2\bar\rho\tau^\circ\tilde{\eth}'\tilde{\eth}' f^\circ - \frac{\rho^2}{\bar\rho}\Psi^\circ\tilde{\eth}' f^\circ
+ \bar\rho\bar\Psi^\circ\tilde{\eth}' f^\circ - 6\bar\rho\Omega^\circ\rho'^\circ\tilde{\eth}' f^\circ + 4\rho\bar\rho\tau^\circ\bar\tau^\circ\tilde{\eth}' f^\circ + 8\rho\rho'^\circ\bar\tau^\circ f^\circ \tag{19}
$$

$$
\mathcal{L}_\zeta g_{ab} m^a m^b = -2\bar\rho\tilde{\eth}\tilde{\eth}\xi_l^\circ + 2\tilde{\eth}'\tilde{\eth} f^\circ + 2\Omega^\circ\tilde{\text{Þ}}' f^\circ + 4\bar\rho\tau^\circ\tilde{\eth}' f^\circ + 8\rho'^\circ f^\circ \tag{20}
$$

III. SPHROIDAL EXPANSIONS

$$
\psi_4^{1\circ} = \sum_{\ell m}\int \mathrm{d}\omega\,\psi_{4,\ell m}^{1\circ}(\omega)\,{}_{-2}S_{\ell m}(\theta,\varphi_\star;a\omega)e^{-i\omega u} \tag{21}
$$

$$
\psi_0^{5\circ} = \sum_{\ell m}\int \mathrm{d}\omega\,\psi_{0,\ell m}^{5\circ}(\omega)\,{}_{+2}S_{\ell m}(\theta,\varphi_\star;a\omega)e^{-i\omega u} \tag{22}
$$

$$
\psi_{4,\ell m}^{1\circ}(\omega) = \frac{\omega^4}{{}_{-2}B_{\ell m}(a\omega)\,{}_{-2}\bar B_{\ell -m}(-a\omega) + 9M^2\omega^2}
\left[{}_{-2}\bar B_{\ell -m}(-a\omega)\psi_{0,\ell m}^{5\circ}(\omega) + 3i\omega(-1)^m\bar\psi_{0,\ell -m}^{5\circ}(-\omega)\right] \tag{23}
$$

$$\bar{h}^{1\circ}_{mm} = \sum_{\ell m} \int \mathrm{d}\omega \frac{-2\omega^2}{{}_{-2}B_{\ell m}(a\omega)\, {}_{-2}\bar{B}_{\ell -m}(-a\omega) + 9M^2\omega^2} \times$$
$$\left[{}_{-2}\bar{B}_{\ell -m}(-a\omega)\psi^{5\circ}_{0,\ell m}(\omega) + 3i\omega(-1)^m \bar{\psi}^{5\circ}_{0,\ell -m}(-\omega)\right] {}_{+2}S_{\ell m}(\theta, \varphi_\star; a\omega) e^{-i\omega u} \quad (24)$$

$$f^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{2i\omega^3}{{}_{-2}B_{\ell m}(a\omega)\, {}_{-2}\bar{B}_{\ell -m}(-a\omega) + 9M^2\omega^2} \psi^{5\circ}_{0,\ell m}(\omega)\, {}_{+2}S_{\ell m}(\theta, \varphi_\star; a\omega) e^{-i\omega u} \quad (25)$$

$$e^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{-4\omega^4(-1)^m}{{}_{-2}B_{\ell m}(a\omega)\, {}_{-2}\bar{B}_{\ell -m}(-a\omega) + 9M^2\omega^2} \bar{\psi}^{5\circ}_{0,\ell -m}(-\omega)\, {}_{-2}S_{\ell m}(\theta, \varphi_\star; a\omega) e^{-i\omega u} \quad (26)$$

IV. SCHWARZSCHILD CASE

$${}_{-2}B_{\ell -m}(a\omega) = {}_{2}B_\ell = \frac{1}{4}\frac{(\ell+2)!}{(\ell-2)!} = \frac{1}{4}(\ell+2)(\ell+1)\ell(\ell-1) \quad (27)$$

$${}_{-1}B_{\ell -m}(a\omega) = {}_{1}B_\ell = \frac{1}{2}\frac{(\ell+1)!}{(\ell-1)!} = \frac{1}{2}\ell(\ell+1) \quad (28)$$

$$\Phi^{0\circ} = d^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{-2/3 i\omega(-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} \left[{}_2B_\ell\, {}_1B_\ell - 3iM\omega(1 + \sqrt{{}_2B_\ell})\right] \bar{\psi}^{5\circ}_{0,\ell -m}(-\omega)\, {}_{-2}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \quad (29)$$

$$\Phi^{1\circ} = \sum_{\ell m} \int \mathrm{d}\omega \frac{2\omega^2(-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} \left[{}_2B_\ell - 3iM\omega\right] \bar{\psi}^{5\circ}_{0,\ell -m}(-\omega)\, {}_{-2}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \quad (30)$$

$$\Phi^{2\circ} = \sum_{\ell m} \int \mathrm{d}\omega \frac{4i\omega^3\, {}_2B_\ell / {}_1B_\ell (-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} \bar{\psi}^{5\circ}_{0,\ell -m}(-\omega)\, {}_{-2}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \quad (31)$$

$$\Phi^{3\circ} = e^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{-4\omega^4 (-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} \bar{\psi}^{5\circ}_{0,\ell -m}(-\omega)\, {}_{-2}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \tag{32}$$

$$\zeta_m^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{-2i\omega^3 \sqrt{{}_2B_\ell / {}_1B_\ell}}{{}_2B_\ell^2 + 9M^2\omega^2} \psi^{5\circ}_{0,\ell m}(\omega)\, {}_{+1}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \tag{33}$$

$$\zeta_l^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{\omega^2 \sqrt{{}_2B_\ell}}{{}_2B_\ell^2 + 9M^2\omega^2} \psi^{5\circ}_{0,\ell m}(\omega)\, {}_0Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} + cc. \tag{34}$$

$$\zeta_n^\circ = \sum_{\ell m} \int \mathrm{d}\omega \frac{-\omega^2 \sqrt{{}_2B_\ell}}{{}_2B_\ell^2 + 9M^2\omega^2} \left({}_1B_\ell + \frac{1}{2} \right) \psi^{5\circ}_{0,\ell m}(\omega)\, {}_0Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} + cc. \tag{35}$$

[1] S. Hollands and V. Toomani, Metric reconstruction in Kerr spacetime, Class. Quant. Grav. **43**, 055001 (2026), arXiv:2405.18604 [gr-qc].

[2] S. R. Green, S. Hollands, and P. Zimmerman, Teukolsky formalism for nonlinear Kerr perturbations, Class. Quant. Grav. **37**, 075001 (2020), arXiv:1908.09095 [gr-qc].

[3] V. Toomani, P. Zimmerman, A. Spiers, S. Hollands, A. Pound, and S. R. Green, New metric reconstruction scheme for gravitational self-force calculations, Class. Quant. Grav. **39**, 015019 (2022), arXiv:2108.04273 [gr-qc].